

SOFTWARE ENGINEERING PROJECT

ACADEMIC CALENDAR

GROUP

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Requirements Specification

Project Objective

The primary objective of the Shared Academic Calendar Mobile Application is to provide a single, unified, and highly accessible scheduling platform for the university community. The system aims to significantly improve operational efficiency and student engagement by ensuring that all academic stakeholders have instant, accurate, and personalized access to course schedules, lecture timings, lab sessions, and critical deadlines via their mobile devices (Android and iOS).

Project Scope

The system will be developed as a cross-platform mobile application utilizing a single codebase for both Android and iOS environments.

In-Scope:

- **Mobile Access:** Native app functionality (Android/iOS).
- **User Roles:** Student, Lecturer, and Admin staff roles (role-based permissions).
- **Event Management:** CRUD operations for Lecturers/Admins (schedule, edit, cancel).
- **Core Functions:** Notifications, personalized schedule views, and **offline data caching**.

Defining the exact stack is crucial for managing development, deployment, and future maintenance.

Area	Scope Inclusion	Detail
Frontend Platform	Cross-Platform Mobile App	The user interface will be built using React Native . This ensures a single codebase for deployment to both the Android and iOS platforms.
Backend API	Server-Side Logic	The core business logic and RESTful API endpoints will be developed using Java with the Spring Boot framework.
Database & Realtime	Data Management & Services	The database will be PostgreSQL , hosted and managed via Supabase . This includes leveraging Supabase services for Realtime updates (for in-app notifications) and potentially Edge Functions (for push notifications).

Functional Requirements (Mobile App Features)

Focusing on the 'Must-Have' features for a mobile environment:

Feature Area	Description	Associated Actor(s)
User Auth	The app will facilitate secure login via institutional credentials.	All Users
User Profile	The app will automatically display events relevant to the user's registered courses/department.	Student, Lecturer
Event View	The app will provide a seamless 'Today's Schedule' view upon launch.	All Users
Schedule	The app allows Lecturers to schedule new events, requiring input for <i>Course</i> , <i>Date/Time</i> , and <i>Location</i> .	Lecturer, Admin
Recurrence	The scheduling interface allow setting up recurring events (e.g., Weekly on Tuesday).	Lecturer, Admin
Notifications	The app uses native push notifications to alert users of upcoming events and critical changes/cancellations.	All Users
Event Detail	The event detail screen includes a dedicated section for event notes/announcements posted by the Lecturer.	All Users

Non-Functional Requirements

Non-Functional Requirements define how the Shared Academic Calendar Mobile App must perform, ensuring a high-quality, reliable, and secure user experience, independent of the core features (like scheduling or viewing).

Category	Requirement Statement
Performance	Offline Access: Users able to view their schedule without network connectivity.
Speed	App synchronization of new data want to be fast.
Usability	The app use native UI conventions for platform consistency.
Security	The app will allow local data access to be protected.